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SELFCHECKGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models

Potsawee Manakul, Adian Liusie, Mark J. F. Gales
ALTA Institute, Department of Engineering, University of Cambridge
pm574@cam.ac.uk, al826@cam.ac.uk, mjfg@eng.cam.ac.uk

Abstract

Generative Large Language Models (LLMs) such as GPT-3 are capable of generating highly fluent responses to a wide variety of user prompts. However, LLMs are known to hallucinate facts and make non-factual statements which can undermine trust in their output. Existing fact-checking approaches either require access to the output probability distribution (which may not be available for systems such as ChatGPT) or external databases that are interfaced via separate, often complex, modules. In this work, we propose "SelfCheckGPT", a simple sampling-based approach that can be used to fact-check the responses of black-box models in a zero-resource fashion, i.e. without an external database. SelfCheckGPT leverages the simple idea that if an LLM has knowledge of a given concept, sampled responses are likely to be similar and contain consistent facts. However, for hallucinated facts, stochastically sampled responses are likely to diverge and contradict one another. We investigate this approach by using GPT-3 to generate passages about individuals from the WikiBio dataset, and manually annotate the factuality of the generated passages. We demonstrate that SelfCheckGPT can: i) detect non-factual and factual sentences; and ii) rank passages in terms of factuality. We compare our approach to several baselines and show that our approach has considerably higher AUC-PR scores in sentence-level hallucination detection and higher correlation scores in passage-level factuality assessment compared to grey-box methods.[1]

1 Introduction

Large Language Models (LLMs) such as GPT-3 (Brown et al., 2020) and PaLM (Chowdhery et al., 2022) are capable of generating fluent and realistic responses to a variety of user prompts. They have been used in many applications such as automatic tools to draft reports, virtual assistants and summarization systems. Despite the convincing and realistic nature of LLM-generated texts, a growing concern with LLMs is their tendency to hallucinate facts. It has been widely observed that models can confidently generate fictitious information, and worryingly there are few, if any, existing approaches to suitably identify LLM hallucinations.

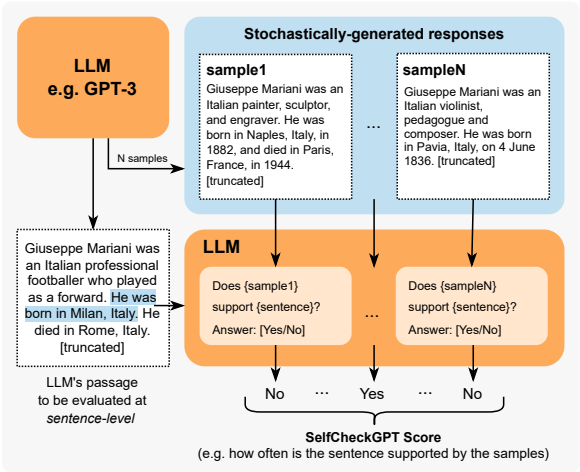

Figure 1: SelfCheckGPT with Prompt. Each LLM-generated sentence is compared against stochastically generated responses with no external database. A comparison method can be, for example, through LLM prompting as shown above.

A possible approach of hallucination detection is to leverage existing intrinsic uncertainty metrics to determine the parts of the output sequence that the system is least certain of (Yuan et al., 2021; Fu et al., 2023). However, uncertainty metrics such as token probability or entropy require access to token-level probability distributions, information which may not be available to users for example when systems are accessed through limited external APIs. An alternate approach is to leverage fact-verification approaches, where evidence is retrieved from an external database to assess the veracity of a claim (Thorne et al., 2018; Guo et al., 2022). However, facts can only be assessed relative to the knowledge present in the database. Addition-

[1] Code and dataset can be found on the project page at https://github.com/potsawee/selfcheckgpt.

ally, hallucinations are observed over a wide range of tasks beyond pure fact verification (Kryscinski et al., 2020; Maynez et al., 2020).

In this paper, we propose SelfCheckGPT, a sampling-based approach that can detect whether responses generated by LLMs are hallucinated or factual. To the best of our knowledge, SelfCheckGPT is the first work to analyze model hallucination of general LLM responses, and is the first zero-resource hallucination detection solution that can be applied to black-box systems. The motivating idea of SelfCheckGPT is that when an LLM has been trained on a given concept, the sampled responses are likely to be similar and contain consistent facts. However, for hallucinated facts, stochastically sampled responses are likely to diverge and may contradict one another. By sampling multiple responses from an LLM, one can measure information consistency between the different responses and determine if statements are factual or hallucinated. Since SelfCheckGPT only leverages sampled responses, it has the added benefit that it can be used for black-box models, and it requires no external database. Five variants of SelfCheckGPT for measuring informational consistency are considered: BERTScore, question-answering, n-gram, NLI, and LLM prompting. Through analysis of annotated articles generated by GPT-3, we show that SelfCheckGPT is a highly effective hallucination detection method that can even outperform grey-box methods, and serves as a strong first baseline for an increasingly important problem of LLMs.

2 Background and Related Work

2.1 Hallucination of Large Language Models

Hallucination has been studied in text generation tasks, including summarization (Huang et al., 2021) and dialogue generation (Shuster et al., 2021), as well as in a variety of other natural language generation tasks (Ji et al., 2023). Self-consistency decoding has shown to improve chain-of-thought prompting performance on complex reasoning tasks (Wang et al., 2023). Further, Liu et al. (2022) introduce a hallucination detection dataset, however, texts are obtained by perturbing factual texts and thus may not reflect true LLM hallucination.

Recently, Azaria and Mitchell (2023) trained a multi-layer perception classifier where an LLM's hidden representations are used as inputs to predict the truthfulness of a sentence. However, this approach is a white-box approach that uses the internal states of the LLM, which may not be available through API calls, and requires labelled data for supervised training. Another recent approach is self-evaluation (Kadavath et al., 2022), where an LLM is prompted to evaluate its previous prediction, e.g., to predict the probability that its generated response/answer is true.

2.2 Sequence Level Uncertainty Estimation

Token probabilities have been used as an indication of model certainty. For example, OpenAI's GPT-3 web interface allows users to display token probabilities (as shown in Figure 2), and further uncertainty estimation approaches based on aleatoric and epistemic uncertainty have been studied for autoregressive generation (Xiao and Wang, 2021; Malinin and Gales, 2021). Additionally, conditional language model scores have been used to evaluate properties of texts (Yuan et al., 2021; Fu et al., 2023). Recently, semantic uncertainty has been proposed to address uncertainty in free-form generation tasks where probabilities are attached to concepts instead of tokens (Kuhn et al., 2023).

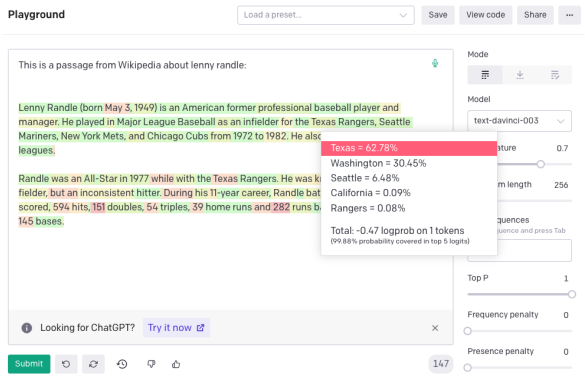

Figure 2: Example of OpenAI's GPT-3 web interface with output token-level probabilities displayed.

2.3 Fact Verification

Existing fact-verification approaches follow a multi-stage pipeline of claim detection, evidence retrieval and verdict prediction (Guo et al., 2022; Zhong et al., 2020). Such methods, however, require access to external databases and can have considerable inference costs.

3 Grey-Box Factuality Assessment

This section will introduce methods that can be used to determine the factuality of LLM responses in a zero-resource setting when one has full access

to output distributions.[2] We will use 'factual' to define when statements are grounded in valid information, i.e. when hallucinations are avoided, and 'zero-resource' when no external database is used.

3.1 Uncertainty-based Assessment

To understand how the factuality of a generated response can be determined in a zero-resource setting, we consider LLM pre-training. During pre-training, the model is trained with next-word prediction over massive corpora of textual data. This gives the model a strong understanding of language (Jawahar et al., 2019; Raffel et al., 2020), powerful contextual reasoning (Zhang et al., 2020), as well as world knowledge (Liusie et al., 2023). Consider the input "`Lionel Messi is a _`". Since Messi is a world-famous athlete who may have appeared multiple times in pre-training, the LLM is likely to know who Messi is. Therefore given the context, the token "`footballer`" may be assigned a high probability while other professions such as "`carpenter`" may be considered improbable. However, for a different input such as "`John Smith is a _`", the system will be unsure of the continuation which may result in a flat probability distribution. During inference, this is likely to lead to a non-factual word being generated.

This insight allows us to understand the connection between uncertainty metrics and factuality. Factual sentences are likely to contain tokens with higher likelihood and lower entropy, while hallucinations are likely to come from positions with flat probability distributions with high uncertainty.

Token-level Probability

Given the LLM's response R, let i denote the i-th sentence in R, j denote the j-th token in the i-th sentence, J is the number of tokens in the sentence, and p_{ij} be the probability of the word generated by the LLM at the j-th token of the i-th sentence. Two probability metrics are used:

$$\text{Avg}(-\log p) = -\frac{1}{J}\sum_{j} \log p_{ij}$$

$$\text{Max}(-\log p) = \max_{j} \left(-\log p_{ij}\right)$$

$\text{Max}(-\log p)$ measures the sentence's likelihood by assessing the *least* likely token in the sentence.

[2]Alternate white-box approaches such as that of Azaria and Mitchell (2023) require access to full internal states, and is less practical and so not considered in this work.

Entropy

The entropy of the output distribution is:

$$\mathcal{H}_{ij} = -\sum_{\tilde{w}\in\mathcal{W}} p_{ij}(\tilde{w}) \log p_{ij}(\tilde{w})$$

where $p_{ij}(\tilde{w})$ is the probability of the word $\tilde{w}$ being generated at the j-th token of the i-th sentence, and $\mathcal{W}$ is the set of all possible words in the vocabulary. Similar to the probability-based metrics, two entropy-based metrics are used:

$$\text{Avg}(\mathcal{H}) = \frac{1}{J}\sum_{j} \mathcal{H}_{ij}; \quad \text{Max}(\mathcal{H}) = \max_{j} \left(\mathcal{H}_{ij}\right)$$

4 Black-Box Factuality Assessment

A drawback of grey-box methods is that they require output token-level probabilities. Though this may seem a reasonable requirement, for massive LLMs only available through limited API calls, such token-level information may not be available (such as with ChatGPT). Therefore, we consider black-box approaches which remain applicable even when only text-based responses are available.

Proxy LLMs

A simple approach to approximate the grey-box approaches is by using a proxy LLM, i.e. another LLM that we have full access to, such as LLaMA (Touvron et al., 2023). A proxy LLM can be used to approximate the output token-level probabilities of the black-box LLM generating the text. In the next section, we propose SelfCheckGPT, which is also a black-box approach.

5 SelfCheckGPT

SelfCheckGPT is our proposed black-box zero-resource hallucination detection scheme, which operates by comparing multiple sampled responses and measuring consistency.

Notation: Let R refer to an LLM response drawn from a given user query. SelfCheckGPT draws a further N stochastic LLM response samples $\{S^1, S^2, ..., S^n, ..., S^N\}$ using the same query, and then measures the consistency between the response and the stochastic samples. We design SelfCheckGPT to predict the hallucination score of the i-th sentence, $\mathcal{S}(i)$, such that $\mathcal{S}(i) \in [0.0, 1.0]$, where $\mathcal{S}(i) \rightarrow 0.0$ if the i-th sentence is grounded in valid information and $\mathcal{S}(i) \rightarrow 1.0$ if the i-th sen-